

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Ethical Hacking

COURSE CODE: ITA14

COURSE OUTCOME COVERED

CO2: Foot printing Tools –: Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT4)

Assessment Tool Weightage: 30%

Duration: 90 min

Total Marks: 50

Scenario-Based Assessment

Topic 1: Footprinting Tools – Conducting Competitive Intelligence

Q1. (L3 – Apply) – 15 Marks

A cybersecurity consultant has been assigned to perform a passive security assessment for a company named ABC Technologies. The consultant must collect publicly available information without directly interacting with the company's servers.

Task: Apply suitable footprinting tools to identify the company's:

- domain registration details,
- employee email addresses,
- technology stack,
- and publicly exposed subdomains.

Explain how each selected tool helps in conducting competitive intelligence.

COMMANDS:

dig abctechnologies.com

dig NS abctechnologies.com

theHarvester -d abctechnologies.com -b crtsh

subfinder -d abctechnologies.com -silent

whatweb <https://abctechnologies.com>

OUTPUTS:

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$$\lfloor (\text{ramu2005} \oplus \text{ramu}) - [\sim] \rfloor$$

```
$ theHarvester -d abctechnologies.com -b crtsh
```

```
Read proxies.yaml from /etc/theHarvester/proxies.yaml
```

[illegible]

* 1 2 3 4 5 6 7 8 9 10 11 12 *

* | | | | | _ / / - / (| | \ W / - ^ _ \ || _ / | *

* _ | _ | _ | v / _ , _ | _ | _ | _ | _ | *

```
* theHarvester 4.10.1 *
```

* Coded by Christian Mantovella

* coded by Christian Martorella
* Elio Geronzi, Research

* Edge-Security Research *

```
* cmartorella@edge-security.com *
```

[*] Target: abctechnologies.com

[*] Searching CRTsh.

```
[*] No IPs found.
```

```
[*] No emails found.
```

```
[*] No people found.
```

```
[*] No hosts found.
```

$$\lfloor (\text{ramu2005} \oplus \text{ramu}) - [2] \rfloor$$

```
$ subfinder -d abctechnologies.com -silent
```

Command 'subfinder' not found, but can be installed with:

```
sudo apt install subfinder
```

Do you want to install it? (N/y)y

```
sudo apt install subfinder
```

```
[sudo] password for ramu2005:
```

```
Installing:
```

subfinder

Summary:

```
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 788
```

Download size: 8,200 kB

Space needed: 32.5 MB / 2,891 MB available

```
Get:1 http://kali.download/kali kali-rolling/main amd64 subfinder amd64 2.14.0-0kali1 [8,200
```

Fetches 8,200 kB in 1s (5,512 kB/s)

```
Selecting previously unselected package subfinder.
```

```
(Reading database... 424761 files and directories currently installed.)
```

```
Preparing to unpack .../subfinder_2.14.0-0kali1_amd64.deb...
```

```
zsh: corrupt history file /home/ramu2005/.zsh_history
```

```
(ramu2005@ramu)-[~]
```

```
$ dig abctechnologies.com
```

```
;; communications error to 8.8.8.8#53: timed out
```

```
; <<>> DiG 9.20.23-1-Debian <<>> abctechnologies.com
```

```
;; global options: +cmd
```

```
;; Got answer:
```

```
;; -->HEADER<-- opcode: QUERY, status: NOERROR, id: 2381
```

```
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
```

```
;; OPT PSEUDOSECTION:
```

```
; EDNS: version: 0, flags;; udp: 512
```

```
;; QUESTION SECTION:
```

```
;abctechnologies.com.          IN      A
```

```
;; ANSWER SECTION:
```

```
abctechnologies.com.    3600    IN      A      18.224.83.222
```

```
;; Query time: 272 msec
```

```
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
```

```
;; WHEN: Sat Aug 08 14:22:53 IST 2026
```

```
;; MSG SIZE rcvd: 64
```

```
(ramu2005@ramu)-[~]
```

```
$ dig NS abctechnologies.com
```

```
; <<>> DiG 9.20.23-1-Debian <<>> NS abctechnologies.com
```

```
;; global options: +cmd
```

```
;; Got answer:
```

```
;; -->HEADER<-- opcode: QUERY, status: NXDOMAIN, id: 13379
```

```
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1
```

```
;; OPT PSEUDOSECTION:
```

```
; EDNS: version: 0, flags;; udp: 512
```

```
;; QUESTION SECTION:
```

```
;abctechnologies.com.          IN      NS
```

```
;; AUTHORITY SECTION:
```

```
com.          900     IN      SOA     a.gtld-servers.net. nstld.verisign-grs.com.
```

```
;; Query time: 48 msec
```

```
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
```

```
;; WHEN: Sat Aug 08 14:23:14 IST 2026
```

```
;; MSG SIZE rcvd: 120
```

```
(ramu2005@ramu)-[~]
```

```
$ theHarvester -d abctechnologies.com -b crtsh
```

```
Read proxies.yaml from /etc/theHarvester/proxies.yaml
```

[*] No hosts found.

Home

(ramu2005@ramu)-[~]

\$ subfinder -d abctechnologies.com -silent

Command 'subfinder' not found, but can be installed with:

sudo apt install subfinder

Do you want to install it? (N/y)y

sudo apt install subfinder

[sudo] password for ramu2005:

Installing:

subfinder

Summary:

Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 788

Download size: 8,200 kB

Space needed: 32.5 MB / 2,891 MB available

Get:1 http://kali.download/kali kali-rolling/main amd64 subfinder amd64 2.14.0-0kali1 [8,200

kB] Fetched 8,200 kB in 1s (5,512 kB/s)

Selecting previously unselected package subfinder.

(Reading database... 424761 files and directories currently installed.)

Preparing to unpack .../subfinder_2.14.0-0kali1_amd64.deb...

Unpacking subfinder (2.14.0-0kali1)...

Setting up subfinder (2.14.0-0kali1)...

Processing triggers for kali-menu (2026.2.6)...

(ramu2005@ramu)-[~]

\$ subfinder -d abctechnologies.com -silent

ww35.abctechnologies.com

www.abctechnologies.com

ww11.abctechnologies.com

ww17.abctechnologies.com

(ramu2005@ramu)-[~]

\$ whatweb https://abctechnologies.com

^C/usr/lib/ruby/vendor_ruby/whatweb/scan.rb:78:in `alive?': Interrupt

from /usr/lib/ruby/vendor_ruby/whatweb/scan.rb:78:in `block (2 levels) in scan'

from /usr/lib/ruby/vendor_ruby/whatweb/scan.rb:78:in `map'

from /usr/lib/ruby/vendor_ruby/whatweb/scan.rb:78:in `block in scan'

from <internal:kernel>:187:in `loop'

from /usr/lib/ruby/vendor_ruby/whatweb/scan.rb:74:in `scan'

from /usr/bin/whatweb:709:in `<main>'

Topic 2: DNS Zone Transfers

Q2. (L3 – Apply) – 15 Marks

During an authorized penetration test, a security analyst suspects that the target organization's DNS server is misconfigured and may allow unauthorized DNS zone transfers.

Task: Apply the appropriate DNS command or tool to verify whether zone transfers are permitted. If the transfer succeeds, explain how the retrieved DNS records can be used to identify internal hosts, mail servers, and subdomains during reconnaissance.

COMMANDS:

```
dig NS abctecnologies.com
dig NS abctechnologies.com +short
dig AXFR abctechnologies.com @ns1-03.azure-dns.com
dig AXFR abctechnologies.com @ns2-03.azure-dns.net
```

OUTPUTS:

```
(ramu2005@ ramu)-[~]
$ dig NS abctechnologies.com +short
ns4-03.azure-dns.info.
ns3-03.azure-dns.org.
ns2-03.azure-dns.net.
ns1-03.azure-dns.com.

(ramu2005@ ramu)-[~]
$ dig AXFR abctechnologies.com @ns1-03.azure-dns.com

; <<>> DiG 9.20.23-1-Debian <<>> AXFR abctechnologies.com @ns1-03.azure-dns.com
;; global options: +cmd
; Transfer failed.

(ramu2005@ ramu)-[~]
$ dig AXFR abctechnologies.com @ns2-03.azure-dns.net

; <<>> DiG 9.20.23-1-Debian <<>> AXFR abctechnologies.com @ns2-03.azure-dns.net
;; global options: +cmd
; Transfer failed.
```

```
(ramu2005@ramu)-[~]  
$ dig NS abctecnologies.com
```

```
; <<>> DiG 9.20.23-1-Debian <<>> NS abctecnologies.com  
;; global options: +cmd  
;; Got answer:  
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 18809  
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1  
  
;; OPT PSEUDOSECTION:  
; EDNS: version: 0, flags:; udp: 512  
;; QUESTION SECTION:  
;abctecnologies.com.                IN      NS  
  
;; AUTHORITY SECTION:  
com.                                900     IN      SOA     a.gtld-servers.net. nstld.verisign-grs.com. 1  
  
;; Query time: 56 msec  
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)  
;; WHEN: Sat Aug 08 14:34:38 IST 2026  
;; MSG SIZE rcvd: 120
```

Topic 3: Introduction to Social Engineering – Social Engineering Attacks and Countermeasures

Q3. (L3 – Apply) – 20 Marks

Employees of an organization receive an email claiming to be from the IT department, asking them to verify their login credentials through a provided link because of an urgent security update.

Task: Identify the type of social engineering attack and apply suitable countermeasures that employees should follow before responding to the email. Explain how these actions help prevent credential theft.

COMMANDS:

```
dig NS abctechnologies.com
dig MX abctechnologies.com
dig A abctechnologies.com
curl -I https://abctechnologies.com
curl -I -L https://abctechnologies.com
```

OUTPUTS:

```
(ramu2005@ramu)-[~]
$ curl -I https://abctechnologies.com
HTTP/1.1 200 OK
Date: Sat, 08 Aug 2026 09:23:34 GMT
Server: Apache
X-Powered-By: PHP/7.4.21
Vary: Accept-Encoding, Cookie
Cache-Control: max-age=3, must-revalidate
X-Frame-Options: SAMEORIGIN
Strict-Transport-Security: max-age=31536000
Content-Type: text/html; charset=UTF-8
```

```
(ramu2005@ramu)-[~]
$ dig MX abctechnologies.com

; <<>> DiG 9.20.23-1-Debian <<>> MX abctechnologies.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NXDOMAIN, id: 6066
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 512
;; QUESTION SECTION:
;abctechnologies.com.          IN      MX

;; AUTHORITY SECTION:
.                86400   IN      SOA     a.root-servers.net. nstld.verisign-grs.com.

;; Query time: 44 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Sat Aug 08 14:54:17 IST 2026
;; MSG SIZE rcvd: 123
```

```
(ramu2005@ramu)-[~]
$ dig A abctechnologies.com

; <<>> DiG 9.20.23-1-Debian <<>> A abctechnologies.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 47798
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
```



```

(ramu2005@ramu)-[~]
$ dig NS abctechnologies.com

; <<>> DiG 9.20.23-1-Debian <<>> NS abctechnologies.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 64046
;; flags: qr rd ra; QUERY: 1, ANSWER: 4, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 512
;; QUESTION SECTION:
;abctechnologies.com.          IN      NS

;; ANSWER SECTION:
abctechnologies.com.  21600   IN      NS      ns2-03.azure-dns.net.
abctechnologies.com.  21600   IN      NS      ns1-03.azure-dns.com.
abctechnologies.com.  21600   IN      NS      ns3-03.azure-dns.org.
abctechnologies.com.  21600   IN      NS      ns4-03.azure-dns.info.

;; Query time: 52 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Sat Aug 08 14:53:14 IST 2026
;; MSG SIZE rcvd: 182

```

```

(ramu2005@ramu)-[~]
$ dig A abctechnologies.com

; <<>> DiG 9.20.23-1-Debian <<>> A abctechnologies.com
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 47798
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 512
;; QUESTION SECTION:
;abctechnologies.com.          IN      A

;; ANSWER SECTION:
abctechnologies.com.  3453    IN      A      18.224.83.222

;; Query time: 72 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Sat Aug 08 14:54:35 IST 2026
;; MSG SIZE rcvd: 64

```

```

(ramu2005@ramu)-[~]
$ curl -I -L https://abctechnologies.com
HTTP/1.1 200 OK
Date: Sat, 08 Aug 2026 09:25:10 GMT
Server: Apache
X-Powered-By: PHP/7.4.21
Vary: Accept-Encoding, Cookie
Cache-Control: max-age=3, must-revalidate
X-Frame-Options: SAMEORIGIN
Strict-Transport-Security: max-age=31536000
Content-Type: text/html; charset=UTF-8

```

Annexure A

Scenario-Based

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Scenario Based Assessment

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand